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import matplotlib.pyplot as plt
import numpy as np
from scipy.optimize import curve_fit

def expFunc(x, a, b):
    return a*np.exp(-b*x)
def powFunc(x, a, b):
    return a / (x**b)

x = np.linspace(0.25, 2 * np.pi, 100) # starting x at zero causes div
by zero errors in the powFunc
y = expFunc(x, 20, 2.5)

sigma = 1.8 * (1-(x / (2*np.pi))) + 0.2
y = np.random.normal(y, sigma, 100)

expParams, expCov = curve_fit(expFunc, x, y)
powParams, powCov = curve_fit(powFunc, x, y)

expY = expFunc(x, expParams[0], expParams[1])
powY = powFunc(x, powParams[0], powParams[1])

plt.scatter(x, y, color='#f00', label='Data')
plt.plot(x, expY, color='#4e4', label='Exponential Fit')
plt.plot(x, powY, color='#44e', label='Power Fit')
plt.xlabel("x")
plt.ylabel("y")
plt.legend()
plt.show()

```

